

Joint Training of Latent-Space Energy-Based Models and β -VAEs for Anime Face Generation

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Abstract

Generating high-quality images requires generative models that are both expressive and tractable to train. We present a hybrid deep generative model that jointly trains a β -VAE and an energy-based model (EBM) in the VAE latent space, combining the structured representations of VAEs with the distributional precision of EBMs for anime face image synthesis. Unlike sequential approaches that use the EBM solely as an inference-time prior, our formulation introduces a bidirectional coupling: the EBM shapes the VAE posterior during training via an energy penalty, while the VAE provides a structured latent space for efficient sampling. Operating entirely in the 256-dimensional latent space avoids the high cost of pixel-space MCMC. Experiments on the Anime Face Dataset demonstrate that the energy gap between posterior and SGLD-generated samples grows from near-zero to -54.25 over joint training, with no degradation in reconstruction quality or posterior collapse.

Keywords: Energy-Based Models, Variational Autoencoders, Latent Space Modeling, Generative Models, Image Synthesis, Stochastic Gradient Langevin Dynamics, Contrastive Divergence, Deep Learning

1 Introduction

Energy-Based Models (EBMs) define a probability distribution over inputs via a scalar energy function $E_\theta(x)$, where low energy corresponds to high probability [1]. Unlike VAEs [2] or GANs [3], EBMs place no structural constraints on the generator, granting flexibility in representing complex, multimodal distributions at the cost of an intractable normalizing constant. Recent work has overcome many scalability barriers: Du and Mordatch [4] showed MCMC-based EBM training can scale to challenging image benchmarks, and Grathwohl et al. [5] demonstrated that a single EBM can simultaneously achieve strong generative and discriminative performance.

A persistent challenge is that MCMC sampling in pixel space is expensive and mixes slowly. Pang et al. [6] proposed learning an EBM as an expressive prior in the low-dimensional latent space of a generator, where MCMC mixes efficiently. We build on this idea and propose a *joint* training framework combining a β -VAE with a latent-space EBM for anime face generation. Rather than treating the two components as separate

modules, the EBM actively shapes the VAE posterior during training by penalizing high-energy encoded samples, while the VAE provides a structured manifold that makes EBM sampling tractable.

The contributions of this work are as follows:

- A joint VAE and latent-space EBM training framework in which the EBM refines the VAE latent prior via contrastive divergence.
- A two-phase training schedule with annealed KL weighting (β -schedule) and EBM shaping ramp-up (α -schedule) for training stability.
- Evaluation on the Kaggle Anime Face Dataset analyzing energy gap dynamics, reconstruction quality, and qualitative comparison of EBM-guided versus Gaussian sampling.
- Critical discussion of limitations relative to VAEBM [7] and controllable latent EBMs [8].

2 Background and Related Work

2.1 Variational Autoencoders

VAEs [2] learn to encode data into a structured probabilistic latent space by optimizing the ELBO:

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z})) \quad (1)$$

Optimization is enabled by the reparameterization trick: $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$. The β -VAE [9] scales the KL term by $\beta > 1$ to encourage disentangled representations.

2.2 Energy-Based Models and SGLD

EBMs [1] define $p_\theta(\mathbf{x}) = e^{-E_\theta(\mathbf{x})}/Z_\theta$, where Z_θ is intractable. Training proceeds via contrastive divergence, whose gradient decomposes into a positive phase (pushing up energy on data samples) and a negative phase (pushing down energy on model samples) [4]. Negative samples are drawn using SGLD [10]:

$$\mathbf{z}_{t+1} = \mathbf{z}_t - \frac{\eta}{2} \nabla_{\mathbf{z}} E_\psi(\mathbf{z}_t) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \eta) \quad (2)$$

Injecting Gaussian noise prevents collapse to a MAP solution; as $\eta \rightarrow 0$ the iterates converge to samples from the true posterior [10].

2.3 Latent-Space EBMs

Pang et al. [6] replace the Gaussian VAE prior with a learned EBM prior, improving sample quality without modifying the decoder. Xiao et al. [7] propose VAEBM, combining a VAE with a pixel-space EBM trained in two stages; MCMC is accelerated via reparameterization in latent space. Nie et al. [8] place an EBM over a pre-trained StyleGAN latent space, enabling compositional zero-shot generation via ODE-based sampling. Our work differs from Pang et al. by training both components jointly and feeding the EBM energy penalty back into the VAE objective during training, rather than using the EBM solely at inference time.

3 Method

3.1 Architecture

The model consists of three components: a VAE encoder, a VAE decoder, and a lightweight EBM operating in the shared latent space. Figure 1 illustrates the full architecture and the two-step training procedure.

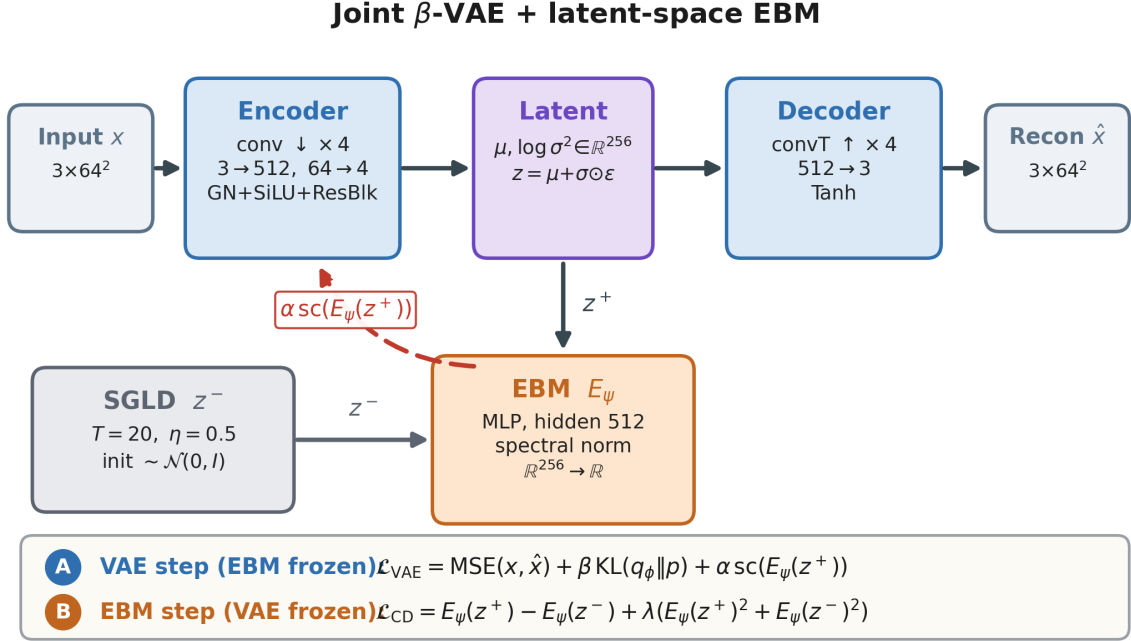


Figure 1: Architecture of the joint β -VAE and latent-space EBM. The encoder maps input images to a 256-dimensional posterior; the EBM receives positive samples $\mathbf{z}^+$ from the encoder and negative samples $\mathbf{z}^-$ from SGLD. The dashed red arrow indicates the energy penalty fed back into the VAE objective during Step A.

The encoder progressively compresses the input image $\mathbf{x} \in \mathbb{R}^{3 \times 64 \times 64}$ into a 256-dimensional Gaussian posterior $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$, with residual connections and GroupNorm stabilizing training at each spatial scale. The decoder mirrors this structure, expanding the latent vector back to image space via transposed convolutions. The EBM is deliberately kept lightweight at 262K parameters (2.6% of total), as its role is discriminative rather than generative. Spectral normalization on all EBM layers bounds the Lipschitz constant of the network, preventing energy collapse during training.

3.2 Training Procedure

Training proceeds in two stages. First, the VAE is pretrained alone for 30 epochs with β annealed from 0 to 1 over 15 epochs, ensuring a well-formed latent space before joint training begins.

In the joint training stage (30 epochs), each iteration alternates two steps:

Step A — VAE update (EBM frozen):

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{MSE}} + \beta \cdot D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) + \alpha \cdot \text{sc}(E_\psi(\mathbf{z}^+)) \quad (3)$$

where $\text{sc}(x) = 10 \cdot \tanh(x/10)$ soft-clamps the energy penalty, and α is linearly ramped from 0 to 1.0 over the first third of joint training to decouple EBM learning from VAE optimization in early stages.

Step B — EBM update (VAE frozen):

$$\mathcal{L}_{\text{CD}} = E_{\psi}(\mathbf{z}^+) - E_{\psi}(\mathbf{z}^-) + \lambda(E_{\psi}(\mathbf{z}^+)^2 + E_{\psi}(\mathbf{z}^-)^2) \quad (4)$$

where $\mathbf{z}^-$ is obtained via SGLD ($T=20$, $\eta=0.5$, noise scale 0.1, grad clip 5.0) initialized from $\mathcal{N}(0, I)$, and $\lambda=10^{-4}$ prevents energy divergence. The VAE is optimized with AdamW at 10^{-5} ; the EBM with Adam at 10^{-4} , both with batch size 64.

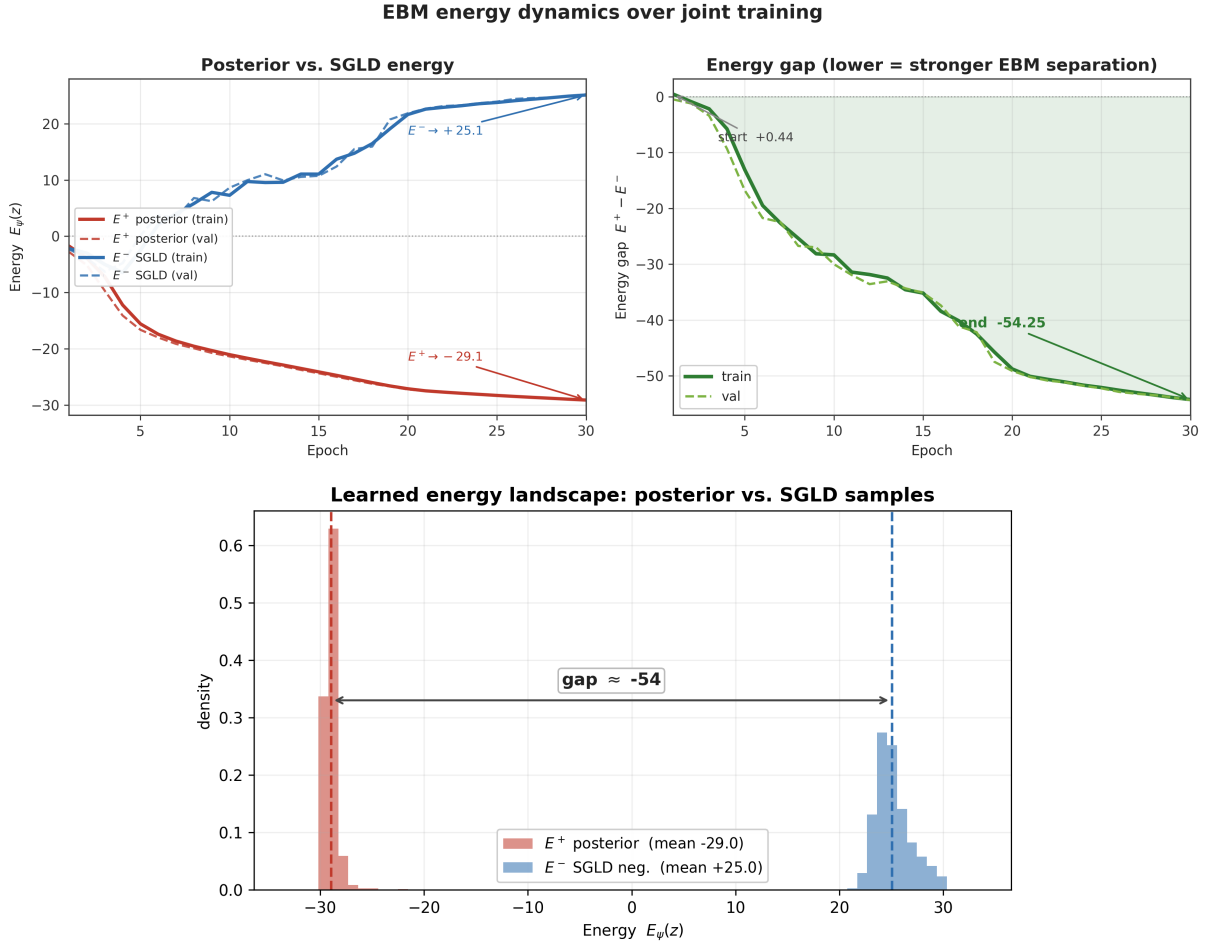


Figure 2: EBM training dynamics. Top: E^+ (posterior) and E^- (SGLD) diverge over training, producing a stable 54-point energy gap. Bottom: at convergence the two distributions are fully separated with no overlap.

4 Experiments

4.1 Setup

We train on the Anime Face Dataset [11], approximately 15,000 images at 64×64 pixels, normalized to $[-1, 1]$.

4.2 Quantitative Results

Figure 2 shows the evolution of E^+ , E^- , and the energy gap over 30 epochs of joint training. The gap grows from near-zero ($+0.44$) to -54.25 , with train and validation curves remaining nearly identical throughout, confirming generalization without overfitting. Reconstruction loss and KL divergence remain stable (train recon: $708.1 \rightarrow 691.1$, KL: $150.9 \rightarrow 141.5$), confirming that the energy penalty does not interfere with VAE optimization and no posterior collapse occurs. SGLD drift remains consistent in the range $[8.27, 9.64]$ with mean 8.67, indicating stable gradient signals throughout the Langevin chain.

4.3 Qualitative Results

Figure 3 shows EBM-guided samples at epoch 5 and epoch 30 alongside Gaussian baseline samples, and Figure 4 shows reconstructions at epoch 30, confirming the VAE retains reconstruction fidelity after joint training. Figure 5 shows linear interpolations between pairs of real images in latent space, demonstrating that the jointly trained VAE learns a smooth and semantically structured manifold, with gradual transitions in hair color, face shape, and other attributes across interpolation steps.

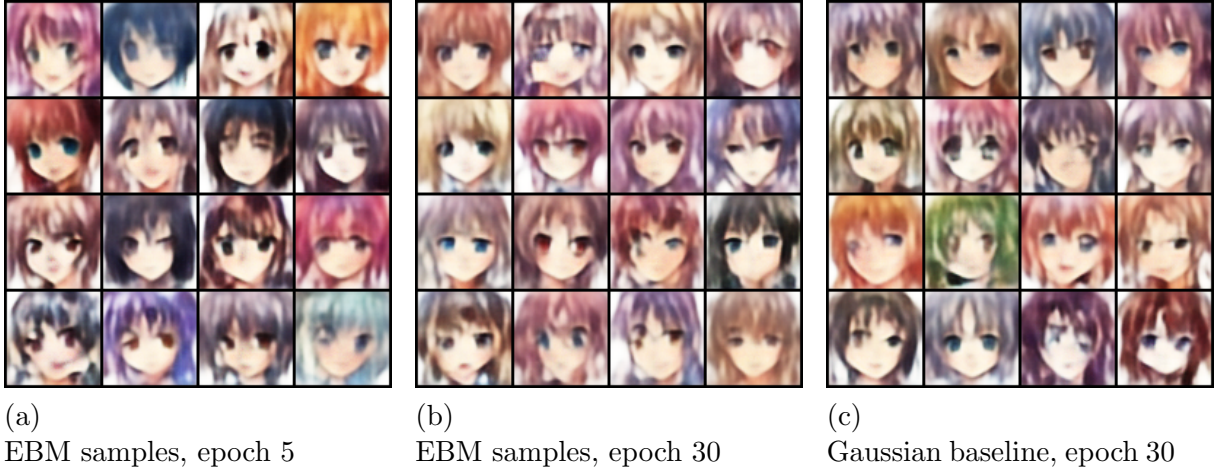


Figure 3: Generated samples. EBM-guided (a,b) via 20-step SGLD; Gaussian baseline (c) decodes $\mathbf{z} \sim \mathcal{N}(0, I)$ directly. Progression from epoch 5 to 30 reflects the widening energy gap.



Figure 4: VAE reconstructions at epoch 30 of joint training.

Latent-space interpolation between real faces (left & right columns are reconstructed endpoints)



Figure 5: Latent space interpolations between pairs of real images. Left and right columns are reconstructed endpoints; intermediate columns are decoded from linearly interpolated latent vectors. Smooth transitions in hair color, face shape, and other attributes indicate a well-structured latent manifold.

5 Discussion

The energy gap evolution from $+0.44$ to -54.25 confirms that the EBM learned a discriminative latent energy landscape without destabilizing the VAE, suggesting the α ramp schedule successfully decouples EBM learning from VAE optimization in early training. Table 1 situates our approach relative to prior work. The key distinction is joint training with bidirectional coupling: unlike all three baselines, the EBM in our model influences the VAE posterior *during* training rather than acting as a post-hoc prior or pixel-space refinement.

Key limitations are threefold. First, the absence of FID evaluation makes quantitative comparison to prior work difficult. Second, the training procedure is sensitive to several hyperparameters (α , β , SGLD step size) determined manually for this setting. Third, fresh-start SGLD at each step is computationally wasteful compared to persistent replay buffers [4]. Future work could address these by incorporating persistent MCMC chains, FID evaluation against a plain VAE baseline, and extension to higher-resolution images via a hierarchical latent space.

Table 1: Comparison to related latent-space EBM approaches.

Method	EBM space	Training	MCMC target	VAE coupling
Pang et al. [6]	Latent	Sequential	Prior $p(\mathbf{z})$	Inference only
VAEBM [7]	Pixel	Sequential	Data $p(\mathbf{x})$	None
Nie et al. [8]	Latent	Sequential	Latent (ODE)	None
Ours	Latent	Joint	Posterior $q_\phi(\mathbf{z} \mathbf{x})$	Bidirectional

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